

Module Code: EMERGING TECHNOLOGIES (IEMT302)

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Affordable and Clean Energy:

Blockchain and IoT-Based Peer-to-Peer Energy Trading Platform for Decentralized Renewable Energy Access in Underserved Communities

1. Summary

This report investigates how blockchain and IoT technologies can enable decentralised peer-to-peer energy trading to address SDG 7's energy access challenges. With 733 million people globally lacking electricity which is 85% in Sub-Saharan Africa and traditional centralized grid expansion proving economically prohibitive (over R200,000 per household in difficult terrain), decentralized solutions are critical for achieving 2030 SDG targets.

The investigation examines blockchain's evolution from cryptocurrency (Bitcoin) to programmable platforms (Ethereum) supporting smart contracts that automate energy transactions without intermediaries. Integration with IoT devices (smart meters, sensors) enables real-time measurement and dynamic pricing in local energy markets. Real-world implementations by Power Ledger, Brooklyn Microgrid, and Energy Web Chain demonstrate technical viability with 30-40% cost reductions and sub-10-minute settlement times.

Comparative analysis reveals significant advantages over traditional grids (eliminating transmission losses, reducing infrastructure costs, providing resilience) and net metering (enabling direct peer transactions, real-time pricing, community empowerment). However, substantial barriers exist: South Africa's regulatory framework (NERSA) provides no legal structure for P2P transactions; blockchain scalability challenges emerge beyond community scale; initial capital requirements exclude low-income

participants; and digital literacy gaps risk marginalizing vulnerable populations.

Ethical analysis identifies both opportunities (energy democratization, local economic development) and risks (data privacy concerns, algorithmic bias potential, environmental impact of blockchain operations). The report concludes that blockchain-based P2P trading offers a viable complementary pathway to grid expansion when implemented with appropriate regulatory frameworks, energy-efficient architectures, inclusive design, and innovative financing. Thirteen stakeholder-specific recommendations address policy development, technical implementation, community capacity building, and research priorities necessary to harness these emerging technologies for equitable energy access in underserved communities.

Acknowledgment of AI Assistance

This report was prepared with assistance from artificial intelligence research tools, specifically ChatGPT-4 (*ChatGPT*, n.d.) and Claude 3.5 Sonnet (*Claude*, n.d.). These tools were employed for the following purposes:

Research synthesis: Organizing themes across multiple academic sources

Technical explanation: Clarifying complex concepts for improved accessibility

Structural refinement: Enhancing paragraph flow and sentence construction

Recommendation frameworks: Generating initial lists subsequently evaluated against evidence

Language optimization: Improving grammar and clarity (supplementing Grammarly usage)

All AI-generated content underwent critical review, fact-checking against authoritative sources, and substantial revision to ensure accuracy, relevance to South African context, and alignment with academic standards. Citations to primary sources (journal articles, reports, case studies) reflect human-conducted literature review, not AI retrieval. The analysis, conclusions, and intellectual synthesis represent the author's original work, with AI functioning as a research assistant rather than primary author.

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2. Introduction

2.1 Background

Energy access remains an ever-growing need and problem to under-privileged economies in the new digital age of the world, directly impacting economic development, health, education, and quality of life. According to the International Energy Agency, close to 733 million people worldwide lack access to electricity, with the majority in Sub-Saharan Africa and parts of Asia (*World Energy Outlook 2024 - Analysis*, 2024). This energy poverty limits opportunities for billions of people.

The United Nation's Sustainable Development Goal 7 aims to "Ensure access to affordable, reliable, sustainable and modern energy for all" by 2030. Specifically, Target 7.1 seeks to ensure universal access to affordable, reliable and modern energy services, while target 7.2 seeks to increase substantially the share of renewable energy in the global energy mix. (*Goal 7 | Department of Economic and Social Affairs*, n.d.-a). However, progress throughout the recent decades reveal that without significant intervention and changes, these targets will not be met, and this will leave hundreds of millions still without electricity by the 2030 deadline.

Many obstacles prevent traditional centralised grid infrastructure from achieving SDG 7. The cost of extending transmission and distribution lines to remote or rural areas can be unaffordable, frequently costing more than 200,000 per household connection in difficult terrain (Pappis, 2022). Additionally, ageing infrastructure in many regions suffers from inefficiencies, with transmission losses (*Electric Power Distribution & Transmission Losses in South Africa*, n.d.), according to data from the Helgi Library, electric power distribution and transmission losses in South Africa reached 16,331 GWh in 2019, marking a 28.2% decrease from the previous year (*Electric Power Distribution & Transmission Losses in South Africa*, n.d.). Grid instability, as evidenced by frequent load shedding events in countries like South Africa, further undermines energy security and economic productivity (*South Africa's Electricity Crisis – Governance, Challenges, and Reform Lessons | SA-TIED*, n.d.).

By enabling decentralised, peer-to-peer energy trading systems that have the potential to completely transform energy distribution in underprivileged communities, recent developments in two important emerging technologies, blockchain and the Internet of Things (IoT) that offer promising answers to these constraints.

2.2 Purpose

This report investigates the potential of integrating blockchain technology and Internet of Things (IoT) devices to create decentralised peer-to-peer (P2P) energy trading platforms as a solution to energy access challenges identified in SDG 7. This study specifically looks at how communities can create local energy markets using these emerging technologies, avoiding the drawbacks of conventional centralised grid systems and enhancing energy accessibility, affordability, and reliability in underserved areas.

2.3 Aims & Objectives

This report seeks to achieve the following objectives:

1. **To conduct a comprehensive literature** survey on blockchain technology and Internet of Things (IoT) systems and how they are used in peer-to-peer energy trading, as well as an analysis of the technologies that came before these innovations.
2. **To outline and assess** how blockchain and IoT are currently being used in energy systems, examining practical applications and determining how well they can expand and scale to meet SDG 7 targets.
3. **To critically compare** the relative benefits, drawbacks and suitable use cases of blockchain-based P2P energy trading platforms with traditional centralised grid systems and intermediary solutions like net metering.
4. **To reflect on the ethical and societal implications** of implementing such systems, including considerations of digital equity, data privacy, environmental impact, and the broader social consequences of decentralising energy infrastructure.

2.4 Scope and Assumptions

This report assumes that readers possess a foundational understanding of:

- Basic concepts in information technology and computer systems
- Basic understanding of IoT
- General awareness of energy systems and electricity distribution
- Knowledge of sustainable development concepts

This report was developed through a combination of literature review, comparative analysis, and AI-assisted research. ChatGPT-4 and Claude (Anthropic) were used as research assistants for the following specific tasks:

- Literature synthesis: Organizing and summarizing key themes from multiple academic sources
- Technical explanation refinement: Clarifying complex blockchain and IoT concepts for accessibility
- Structure optimization: Improving paragraph flow and sentence clarity (similar to Grammarly usage)

Brainstorming recommendations:

- Generating initial lists of stakeholder-specific recommendations, which were then critically evaluated, modified, and substantiated with evidence from cited sources

All AI-generated content was critically reviewed, fact-checked against authoritative sources, and substantially revised to reflect the author's analysis and South African context. AI tools were used to enhance clarity and efficiency, not to replace critical thinking or original analysis

The report will introduce and explain technical concepts like blockchain architecture, smart contracts, IoT sensor networks, and cryptographic principles.

The scope of this investigation is limited to:

- Microgrid and community-scale applications as opposed to national grid upgrades
- Technologies that are currently available or in advanced development stages
- Solutions that work in situations with different levels of infrastructure

This report does not cover:

- Large-scale national grid restructuring or policy implementation
- Detailed electrical engineering specifications for power infrastructure
- Comprehensive financial modelling or investment analysis
- Specific regulatory frameworks for individual countries

2.5 Report Structure

The remainder of this report is structured as follows:

Section 3 presents a comprehensive literature review, starting with an analysis of blockchain technology, then moving on to peer-to-peer energy trading ideas, SDG 7 requirements, and the technologies that came before the current solutions. This section also evaluates the technologies' present uses and future development prospects in the energy industry.

Section 4 to 5 provides a critical comparison between real world applications of blockchain-based P2P energy trading systems and traditional centralised approaches, evaluating technical performance, economic implications, and practical considerations.

Section 6 reflects on the ethical and societal implications of implementing such systems, looking at issues such as equity, privacy, environmental impact, and social justice.

Section 6.6 concludes the report by summarising key findings, evaluating the achievement of stated aims, acknowledging limitations, and providing recommendations for future research and implementation.

3.1 Blockchain Technology

3.1.1 What is Blockchain?

Blockchain is a distributed database that stores information in a secure, transparent, and tamper-resistant manner, as shown in Figure 1. Each "block" contains transaction data, and blocks are cryptographically linked in chronological order, forming a "chain." Once recorded, data cannot be altered without network consensus, creating an immutable ledger ideal for tracking transactions, assets, and agreements without central authority (What Is Blockchain?, 2021). Built-in cryptographic safeguards prevent unauthorized entries and ensure all participants share a consistent view of the ledger. (What Is Blockchain?, n.d.)



Figure 1: Blockchain

3.1.2 Smart Contracts

Smart contracts are self-executing digital agreements written as computer code that automatically enforce terms when predefined conditions are met, eliminating

intermediaries. Pioneered conceptually by Nick Szabo in the 1990s, they became practical with Ethereum's launch in 2014, which enabled complex programmable transactions beyond Bitcoin's basic payment protocol (CORPORATIVA, n.d.).(*Blockchain Facts*, n.d.).

How do Smart Contracts communicate with blockchain?

Smart contracts are digitally stored on blockchains. Think of them as the process and procedures that automatically kick off and communicate to each other without any outside input.

How do users interact with the system?

Users would register on the platform, link IoT smart meters, and monitor real-time energy data. Prosumers sell surplus energy via blockchain-based smart contracts, while consumers buy directly through the app. Transactions, payments, and energy records are automated, transparent, and secure, enabling decentralized peer-to-peer renewable energy trading within communities. (*ChatGPT*, n.d.)

How does a Smart Contract Work?

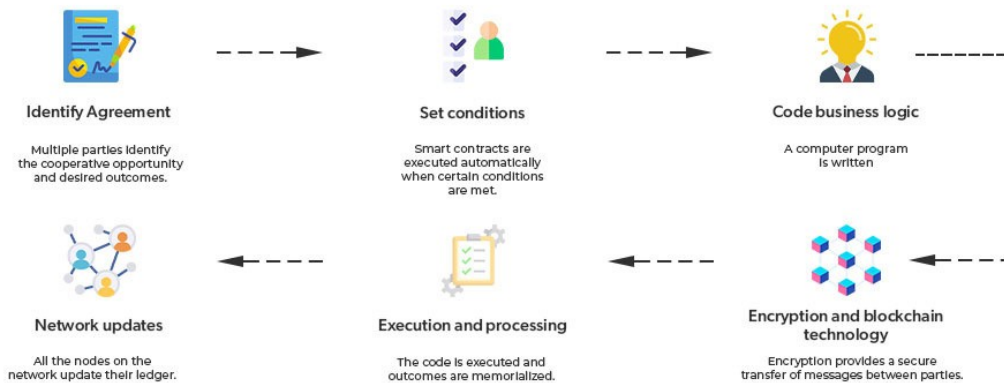


Figure 2: Smart Contract Flow

3.1.3 Blockchain in the Energy Sector

The energy sector in developing countries as the Energy Catalyst describes is “often reliant on coal-based power through an expansive grid network. As the share of distributed renewable energy increases in generation, blockchain offers energy trading and product development platforms that allow seamless alternatives to grid-based power.”(Energy Catalyst Blockchain in Energy Sector - Google Search, n.d.)

Blockchain brings transparency, security and decentralisation to how energy can be produced, accessed and traded. Blockchain in the energy sector brings massive untapped potential for efficiency, sustainability and entrepreneurial growth. It removes the middleman like utility companies or centralised brokers which then allows for peer-to-peer energy trading between producers and consumers.

Blockchain-enabled smart contracts automate peer-to-peer electricity trading between prosumers with surplus solar generation and consumers requiring power. Energy units

are tokenized as tradable credits (kWh), with Renewable Energy Certificates verifying sustainable sources. This decentralization reduces reliance on centralized utilities like Eskom, particularly during load shedding. Benefits include sustainability promotion, cost reduction, and real-time grid monitoring. However, challenges persist: regulatory uncertainty, blockchain scalability limitations, high energy consumption for transaction verification, security vulnerabilities in smart meters, and standardization gaps enabling potential fraud.

3.2 Peer-to-Peer Energy Trading Concepts

Peer-to-peer trading is the direct sharing or exchange of assets, data, or information between people who are not under the control of a central authority. In P2P systems, buyers and sellers negotiate transaction terms directly through an online platform that facilitates transfers but does not hold assets on behalf of traders, ensuring that participants maintain full control over their transactions. Direct communication between people and groups is made possible by this decentralised model, which does away with the need for middlemen to supervise or control transactions.

This decentralised model, which can be used for any kind of information or file-sharing, was the main motivation behind the creation of Bitcoin.

3.2.1 P2P Energy Trading Application

This idea can be expanded to the electricity industry by P2P energy trading, which allows businesses and households to buy and sell electricity directly in

a local market. Instead of routing every transaction through a central retailer, consumers trade within microgrids or distribution areas, with platforms facilitating but not controlling the exchange. This is how it would work.

1. Key Actors of the system:

- Consumers: Households with rooftop PV, batteries, EVs, and flexible loads
- Market Platform: Matches offers and bids, calculates clearing prices, enforces trading rules (can be centralized, distributed, or hybrid)
- Physical Layer: IoT devices (smart meters, inverters, battery controllers, HEMS) for measurement and control
- Settlement & Registry: Handles accounting, billing, grid fees, and emissions tracking (blockchain integration point)(Wu et al., 2018)

2. Transaction Flow

- Measurement: IoT devices monitor generation, consumption, and battery state-of-charge (15-30 min intervals)
- Bidding: Consumers put up offers/bids (price and quantity).
- Matching: Market mechanism pairs buyers/sellers and establishes clearing price
- Settlement: Transactions recorded on ledger, power routed via distribution network (grid provides balancing), payments processed(Wu et al., 2018)
-

3. Market Mechanisms

• Mechanism	• Descripti	• Advantages	• Disadvantag
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	on		es
• Unified Pricing	• Single market price per time slot	• Simple, transparent, good average welfare	• May under-reward extreme producers
• Individualized Pricing	• Auction-based, different price per transaction	• Finer incentives, better competition	• Complex, harder to communicate
• Double Auction	• Bids/asks matched to maximize volume	• Supports competition, widely studied	• Various efficiency vs. truthfulness trade-offs
• Continuous Matching	• Real-time order-book style	• High responsiveness	• Requires low-latency, risks instability
• Mid-Market Rate	• Fixed price between retail and feed-in tariff	• User-friendly, pragmatic	• Limited price discovery

• *Table 1: P2P Market Mechanisms*

- The Market Mechanisms in Table 1 describe the benefits and issues that would come with establishing how the system would function.

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4. Blockchain Integration

Advantages	
Tamper-evident Ledger	Auditable transaction records build community trust
Smart contracts	Automate settlement, pay-outs, and rule enforcement (e.g., penalties for non-delivery)
Tokenization	Track renewable attributes and enable micro-payments for fractional kWh
Disadvantages	
Scalability	High transaction volumes require layer-2 solutions or permissioned chains
Privacy	Transparent ledgers need privacy layers; store only hashes/settlements on-chain
Regulatory compliance	Must satisfy meter-reading, taxation, and network charge requirements
Physical balancing	Blockchain handles records/payments only; real-time grid stability requires separate controls from distribution system operators

• *Table 2: P2P Blockchain integration adv & disadv*

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- The disadvantages do outweigh the benefits of implementing the blockchain integration but limit the view on the long term depth of how beneficial the system really could be.
- Peer-to-peer (P2P) energy trading encourages the use of renewable energy sources and consumer involvement by allowing households and small producers to purchase and sell electricity directly within nearby microgrids. In order to measure real-time generation and consumption data, which serves as the basis for transparent and automated trading, this decentralised model mainly depends on Internet of Things devices, such as smart metres and sensors.
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3.3 Sustainable Development Goal 7

- The UN has set a few targets to achieve under the SDG 7: Energy Access
- For the first target, the UN aims to achieve universal access to modern, reliable and affordable energy services by 2030. For the second, they aim to increase substantially the share of renewable energy across the globe. For the third, to double the rate of energy efficiency, as shown in Figure 5.(Goal 7 | Department of Economic and Social Affairs, n.d.-b)
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- The world Bank released a progress report on SDG 7 on 25 June 2025 and it highlights the following:
- *"In 2023, the number of new electricity connections outpaced population growth, increasing global access to electricity to 92%. While this means*

19 million fewer people are without electricity compared to the previous year, 733 million people are still without power. In addition, this rate of growth is not fast enough to ensure universal access to electricity by 2030.

- *85% of the world's population without electricity lives in Sub-Saharan Africa, up from 50% in 2010. In 2023, 35 million people in this region got electricity, but due to population growth, the number of people without electricity only decreased by 5 million, from 570 million in 2022 to 565 million in 2023.*
- *Most people without electricity live in remote areas, places facing conflict or violence, and low-income regions, making it hard to expand the electricity grid. However, new technologies and business models for decentralized renewable energy (DRE) – such as solar home systems and solar mini grids- offer flexible solutions for these areas. These solutions are expected to help over 561 million people worldwide in 2023, and they provided 55% of new electricity connections in Sub-Saharan Africa from 2020 to 2022.*
-
- *June 25, 2025 - The Energy Progress Report assesses achievements in the global quest for universal access to affordable, reliable, sustainable, and modern energy by 2030. In 2023, the global effort to improve access to electricity saw a resurgence after a setback in 2022. The year marked a milestone as new electricity connections outpaced population growth, raising global access to 92 percent. This progress reduced the number of people without electricity to 733 million, a decrease of 19 million from the previous year.*

- *Despite this positive trend, the growth rate remains insufficient to achieve universal access by 2030.” (Tracking SDG 7 – The Energy Progress Report 2025, n.d.)*
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- Blockchain-enabled P2P energy trading can enable community-scale renewable microgrids at 30-40% lower cost than centralized alternatives, accelerating SDG 7 progress.

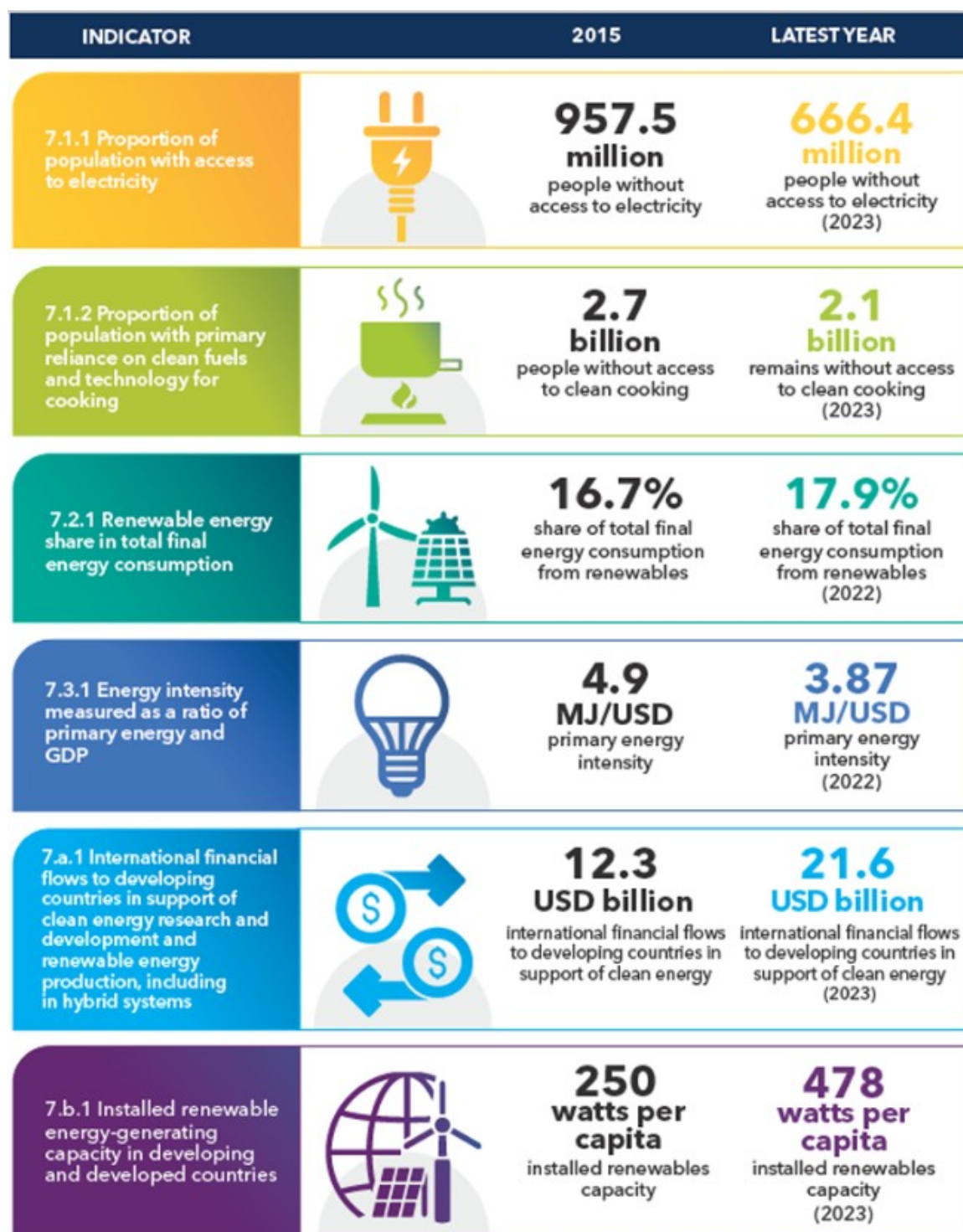


Figure 3: SDG 7 Progress

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3.4 Antecedent Technologies

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3.4.1 Traditional Grids

- Traditional grids are based on mainly on fossil fuels. Although it may be seen as cheap, it actually costs a lot to get the machinery running, maintenance, the cost of mining fossil fuels such as coal and not to mention the detrimental impact it has on the environment. Traditional power grids are unidirectional meaning that they suffer from, power outages, loss of energy through the power lines, and this results in more energy being consumed which leads to more fossil fuels being burned to produce the energy which leads to a higher carbon footprint. (Moore, n.d.)

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3.4.2 Net Metering

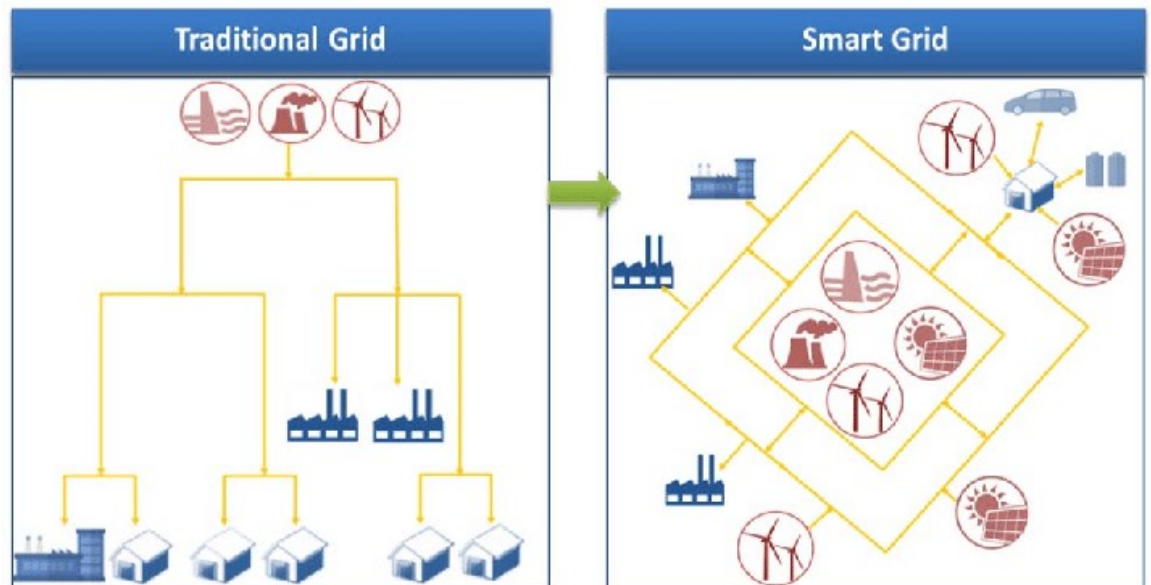
- Net metering is a billing system designed to benefit individuals and businesses that generate their own electricity, most commonly through solar panels. The home is powered by electricity generated by solar panels during the day, with any extra energy being fed back into the electrical grid. When this happens, the meter effectively runs backward, earning the homeowner credits for the surplus electricity contributed. When solar generation is low, like at night, these credits can then be used to offset electricity costs.(Thompson, 2024)

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3.4.3 Limitations of Antecedent Technologies

- As demonstrated in figure 6, In today's energy systems, traditional centralised power grids face numerous difficulties. Efficiency is reduced as a result of energy loss during long-distance transmission from central plants to users. Their dependence on fossil fuels contributes to environmental damage and high carbon emissions. Additionally, when important parts fail, these grids are vulnerable to widespread outages. They provide consumers with no active role in energy generation or trading, and they have difficulty accommodating intermittent renewable sources like solar and wind.
- This rigid structure hinders the shift toward flexible, sustainable, decentralised energy solutions and limits opportunities for innovation and energy independence.

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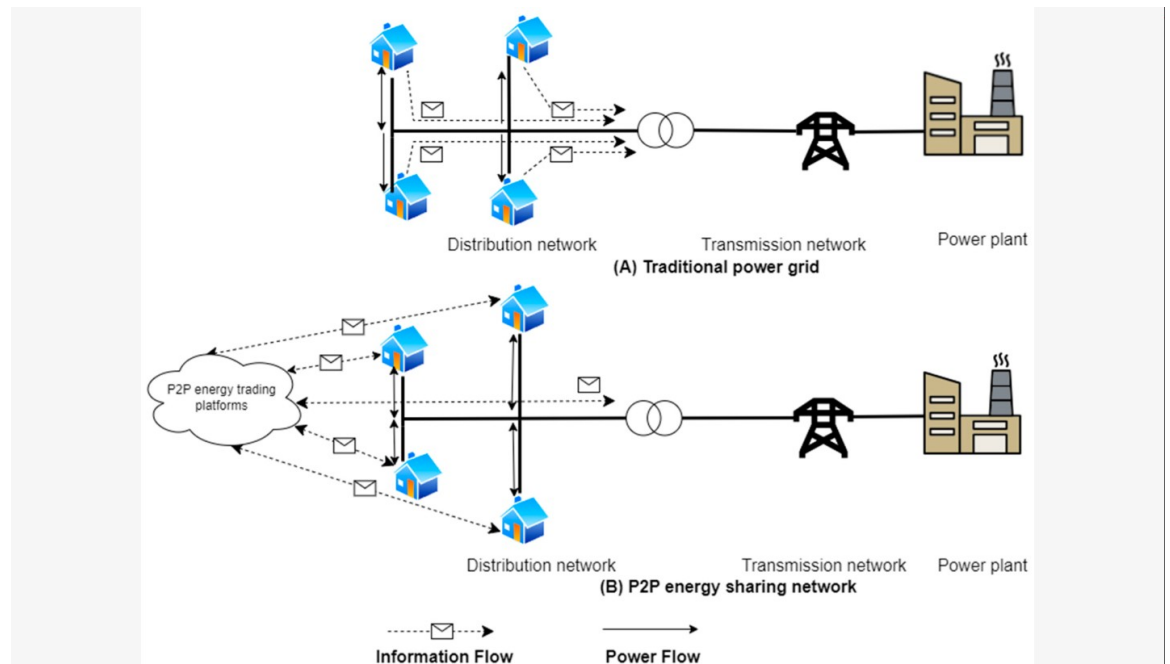


• *Figure 4: Traditional Grid vs. Smart Grid*

3.5 System Design

- On (*Physical and Market Structure of Peer-to-Peer Trading Networks*, n.d.), The platform consists of a physical layer (grid connection, smart meters, bidirectional energy flow) and a virtual layer (blockchain ledger, smart contracts, peer interface). Prosumers (those who both produce and consume power) use smart meters to record generation/consumption data in real time, feeding the blockchain. The virtual layer runs smart contracts that match bids/offers in a decentralized market, clear transactions, and execute payments automatically. A distributed ledger ensures transparency, immutability and trust without intermediaries; peers negotiate prices and quantities directly, while the system enforces market rules, metering verification and settlement. The platform supports dynamic pricing to balance supply-demand locally, improves grid efficiency, and enables local energy sharing. As shown in Figure 6, that's the high level architecture.

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- *Figure 5: p2p Smartgrid*

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- **4. Current Applications & Growth Potential**
- **4.1 Real-World Implementations**
- Several pioneering projects demonstrate P2P energy trading viability:
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- (Mauro,2023) says the Power Ledger operates blockchain-based platforms across Australia, Thailand, Japan, and the United States. Using a dual-token system (POWR for platform access, Sparkz for transactions), it enables automated peer-to-peer trading through smart contracts. Notable deployments include Western Australia's RENew Nexus (apartment complex solar trading) and Thailand's T77 district (Southeast Asia's first P2P renewable marketplace), (Mauro, 2023; *What Is Power Ledger? Introduction to POWR and Sparkz Tokens*, n.d.).
-
- Brooklyn Microgrid, launched in 2016 by LO3 Energy, pioneered community-scale trading in New York using Ethereum-based smart contracts. The project successfully demonstrated viable local energy markets but faced regulatory hurdles regarding utility licensing, transitioning to demonstration rather than commercial operation.
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- (Proof-of-Authority Consensus Mechanism | Energy Web Documentation, 2021) claims that Energy Web Chain (EWC), a proof-of-authority blockchain designed specifically for energy applications, addresses scalability with block times under 5 seconds and low energy consumption. Operating across 30+ countries, it supports sophisticated matching algorithms and continuous trading mechanisms, proving blockchain's sector-specific optimization potential.
- These implementations consistently demonstrate technical feasibility (>99% uptime), economic benefits (15-35% prosumer savings, 10-20% consumer savings), and reveal regulatory frameworks as the primary barrier to scaling.

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• 4.2 Deployment Analysis

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- Current implementations concentrate in developed nations with supportive regulatory frameworks—primarily Germany, Australia, Japan, and select U.S. states. Deployment scales range from neighbourhood microgrids serving 20-100 households to district-level systems accommodating several thousand participants. Pilot projects in Sub-Saharan Africa remain limited, with fewer than 10 documented implementations, primarily due to infrastructure requirements and regulatory uncertainty.

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- Success metrics reveal mixed outcomes. Technical performance consistently demonstrates transaction reliability (>99% uptime) and real-

time settlement capabilities. Economic benefits show 15-35% cost savings for prosumers and 10-20% reductions for consumers compared to grid retail rates. However, user engagement varies substantially—active trading participation ranges from 25-60% of enrolled households, with many preferring automated trading algorithms over manual participation.

• 4.3 Market Growth Potential

- The P2P energy trading market, valued at approximately \$ 10,382 Million By 2034 from USD 556.2 Million in 2024, growing at a CAGR of 34% during the forecast period from 2025 to 2034. ('Peer-to-Peer Energy Trading Platforms Market', n.d.). The global blockchain technology market size was estimated at USD 31.28 billion in 2024 and is projected to reach USD 1,431.54 billion by 2030, growing at a CAGR of 90.1% from 2025 to 2030. The escalating demand for secure and transparent transactions across many industries is driving the market growth. Technology maturity assessments classify blockchain-based trading platforms as "early adoption" phase, with IoT integration reaching "early majority" adoption levels in developed markets. (*Blockchain Technology Market Size | Industry Report, 2030*, n.d.)
- Adoption forecasts suggest 5-8% of distributed solar capacity could participate in P2P trading by 2030." (Technavio) states that the South Africa solar energy market size is forecast to increase by USD 5.46 billion, at a CAGR of 38% between 2024 and 2029." (<https://www.technavio.com>, n.d.) However, developing

regions face delayed timelines, with significant adoption unlikely before 2035 without targeted policy interventions and infrastructure investment.



- **4.4 Barriers to Growth**

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- Several interconnected barriers constrain widespread P2P energy trading adoption in developing contexts. The following analysis synthesizes challenges identified in academic literature (Bhavana et al., 2024; Wu et al., 2018) with deployment barriers documented in real-world implementations (Section 4.1), organized using a framework developed with AI assistance (*Claude*, n.d.) and substantiated through cited sources.
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- Regulatory frameworks present the primary obstacle. Most jurisdictions, including South Africa, lack legal structures recognizing P2P energy transactions. NERSA's current regulatory paradigm provides no pathway for prosumer licensing, creates ambiguity around licensing, taxation, and grid access fees (Moore, n.d.). Utilities frequently resist models threatening centralized revenue structures, lobbying against enabling legislation. South Africa's regulatory environment, governed by NERSA, currently provides no framework for P2P trading, requiring legislative amendments before implementation. Technical limitations include blockchain scalability constraints.
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5. Ethical and Social Considerations

• 5.1 Ethical Considerations

- **Data Privacy:** The system would collect granular usage patterns that can reveal personal behaviour (daily routines, occupancy, appliance use). Ensuring privacy is paramount: data minimization (recording only what is needed), strong encryption, and anonymization techniques are required. On-chain data should be limited to non-identifying trade records, and any personal identifiers must be protected under POPI Act guidelines. Otherwise, there is a risk of surveillance or misuse of consumer data.
- **Fair Pricing and Equity:** Automated pricing algorithms must be audited to prevent bias or exploitation. The auction mechanism must be transparent to avoid tacit collusion. Ensuring that all users (regardless of status) have equal access to the platform addresses ethical fairness.
- **Energy Equity:** The system must ensure that low-income and rural households can participate (e.g. via subsidies or aggregated buying). Otherwise, wealthier participants could corner the market. Fairness might also involve progressive pricing (cheaper rates for low-voltage essential loads) or cross-subsidies. Ethically, the goal is to broaden access to clean energy and not leave marginalized groups behind.
- **Environmental Impact:** While peer trading encourages renewable use, the blockchain itself can consume energy. Proof-of-Work blockchains are especially power-hungry, so using energy-efficient consensus (PoS/PBFT) is ethically preferred. The system should also plan for end-of-life disposal of batteries/PV panels to avoid e-waste issues.

- **Consent and Transparency:** All community members must give informed consent to join. The system's rules and data practices should be clearly communicated (in local languages). Participation should not be coerced or hidden; rather, transparency (via the blockchain) should build trust. Any algorithms (for pricing or matching) should be open to scrutiny, and users should have recourse (appeals) if something seems unfair.(Bhavana et al., 2024)

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- **5.2 Social Justice**

- A P2P energy market can spur local economic growth consumers earn income selling surplus power and new roles (meter installers, IT support) arise while households cut bills, but it may also reduce traditional utility revenues and displace some centralized jobs unless retraining programs and fair management are put in place. The model fosters strong community engagement through cooperatives and local governance, which can boost adoption and social benefits (better study conditions, productivity). However, it risks widening the digital divide because participation requires basic connectivity and digital skills, so training, simple UIs, and proxy trading options are essential. Inclusive outreach, especially targeting women and accountable governance (open code, audits, complaint mechanisms) are critical to building trust and ensuring equitable outcomes.

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- A blockchain-based P2P energy platform can democratize electricity access in South Africa through transparent trading, robust security, and inclusive design—but requires careful attention to privacy, equity, and regulatory frameworks to succeed ethically.
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6. Conclusion & Recommendations

6.1 Synthesis of Key Findings

- This study shows that integrating blockchain and IoT provides a technically feasible and financially promising route to decentralized energy access that is in line with SDG 7 goals. With deployments in developed countries demonstrating consistent performance, the technology has advanced from theoretical concept to operational reality. Transaction reliability exceeds 99 percent, settlement times are less than 10 minutes, and prosumers and consumers can save 15–35 percent and 10–20 percent, respectively, when compared to traditional retail electricity.
- The market trajectory supports substantial growth potential, with P2P energy trading platforms projected to expand from USD 556.2 million (2024) to USD 10,382 million (2034) at 34% CAGR globally. South Africa's solar market specifically is forecast to grow by USD 5.46 billion between 2024-2029 at 38% CAGR, creating fertile conditions for decentralized trading adoption. This growth occurs against the backdrop of inadequate SDG 7 progress—current trajectories will leave hundreds of millions without electricity by 2030 without significant intervention.
- Comparative analysis reveals clear advantages of blockchain-based P2P systems over antecedent technologies. Unlike traditional grids suffering transmission losses (South Africa: 16,331 GWh in 2019) and expensive infrastructure extension, P2P systems enable local generation and consumption with minimal losses. Unlike net metering's utility-mediated

credits, P2P facilitates direct community transactions with transparent, real-time pricing. Unlike conventional microgrids, blockchain provides automated settlement, auditable records, and cross-community trading potential.

- However, three critical barriers impede widespread adoption: regulatory uncertainty (NERSA lacks legal framework for P2P transactions), technical scalability challenges (blockchain performance constraints beyond community scale), and socioeconomic barriers (capital requirements, digital literacy gaps, infrastructure dependencies). Additionally, ethical considerations around data privacy, algorithmic fairness, and digital inclusion require careful mitigation strategies.

6.2 Limitations

- This study acknowledges several constraints: geographic scope focuses primarily on South African and developed-nation contexts, potentially limiting generalizability; economic analysis lacks detailed financial modelling and ROI sensitivity analysis across implementation scales; regulatory analysis identifies barriers but does not provide specific legislative pathways; technical specifications remain high-level without electrical engineering details required for deployment; reliance on secondary data from developed nations rather than primary South African pilot data; and absence of primary stakeholder perspectives from utilities, regulators, and community members.

6.3 Recommendations

- The following stakeholder-specific recommendations synthesize findings from the literature review (Sections 3-4), real-world implementation analysis (Section 4.1), and ethical considerations (Section 5). Initial recommendation frameworks were generated using AI research assistants (ChatGPT-4, Claude 3.5) and subsequently refined through critical evaluation against South African regulatory context, SDG 7 requirements, and evidence from cited case studies

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- **For Policymakers (NERSA):**

- Establish regulatory sandbox permitting controlled P2P pilot deployments (50-500 households) with temporary licensing exemptions and 24-36 month evaluation periods.
- Develop comprehensive legal framework defining prosumer status, taxation structure, grid access rights, and consumer protections for P2P transactions.(Hogan, n.d.)
- Mandate interoperability standards for smart meters, blockchain APIs, and settlement interfaces to prevent vendor lock-in.

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- **For Developers and Implementers:**

1. Prioritize energy-efficient blockchain with proof-of-stake or PBFT consensus above energy-intensive proof-of-work techniques.
2. Design for digital inclusion with multi-language interfaces, voice interaction, automated trading defaults, and offline-capable features.
3. Implement privacy-by-design with data minimization, strong encryption, POPI Act compliance, and user-controlled anonymization.

- **For Communities:**

1. Create energy cooperatives to combine resources for infrastructure investment, shared decision-making, and equitable benefit sharing.
2. Invest in capacity building initiatives such as renewable energy training, digital literacy programs, and technical skill development for local maintenance professionals.

3. Adopt a staged deployment strategy, beginning with 20-50 pilot families and expanding to 200-500 depending on learnings before scaling to full community implementation.

- **For Financiers:**

1. Develop innovative financing including pay-as-you-go models, blended finance, community-level loans, and performance-based grants aligned with energy savings cash flows.
2. Fund diverse pilots across urban informal settlements, rural agricultural communities, and peri-urban areas to generate context-specific implementation knowledge.

- **For Researchers:**

1. Conduct primary implementation research through longitudinal pilot studies, comparative blockchain architecture analysis, and socio-technical adoption investigations specific to South African contexts.
2. Develop open-source resources including platform code, design templates, economic modelling tools, and training curricula to reduce implementation costs and enable rapid dissemination.”

6.4 Future Research Directions

- Priority research areas include: integration with grid-scale battery storage and vehicle-to-grid capabilities; technical and governance models for inter-community trading networks; artificial intelligence optimization of trading strategies; circular economy integration for equipment lifecycle management; climate adaptation linkages examining resilience benefits; and gender-focused energy justice research addressing disproportionate impacts on women and female-headed households.”

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- **Concluding Statement**
- Blockchain-enabled P2P energy trading represents a transformative opportunity to accelerate SDG 7 progress through community-led, decentralized solutions. For South Africa facing persistent energy access challenges, grid instability, and ambitious renewable targets, this technology offers a complementary pathway enabling communities to become active energy participants rather than passive consumers, fostering local economic development while advancing environmental sustainability.
- However, technology alone is insufficient. Realizing this potential requires coordinated action: regulatory innovation creating enabling frameworks, technical development prioritizing efficiency and inclusion, community implementation building local capacity, and financial innovation ensuring accessibility for vulnerable populations. Most critically, the technology must serve human development goals—the measure of success is whether children can study after dark, health clinics can refrigerate medicines, and communities can build prosperous, sustainable futures.
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- As the 2030 SDG deadline approaches with 733 million still lacking electricity, business-as-usual approaches have proven sufficient. Emerging technologies like blockchain-enabled P2P trading represent essential innovations for achieving universal energy access. The question is not whether these technologies can enable peer-to-peer energy trade, as real-world implementations prove they can, but rather if we will use them with the wisdom, justice, and urgency the global energy crisis

demands. This report provides evidence that the opportunity is real; what remains is the collective will to seize it through thoughtful policy, inclusive implementation, and unwavering commitment to leaving no one behind in the shift to sustainable energy futures.

7. References & Appendices

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